

YMDS AI

Bilingual RAG-Based Regulatory Compliance Checking for Higher Education

Problem

Manual YOK compliance review is slow, repetitive, and difficult to audit.

Approach

Article parsing + BM25/Dense/Hybrid/Multi-Query retrieval + GPT reasoning.

System

React bilingual UI, FastAPI backend, PostgreSQL storage, Docker deployment.

Results

RAG improves direct prompting; GPT-4o-mini provides strong cost-performance.

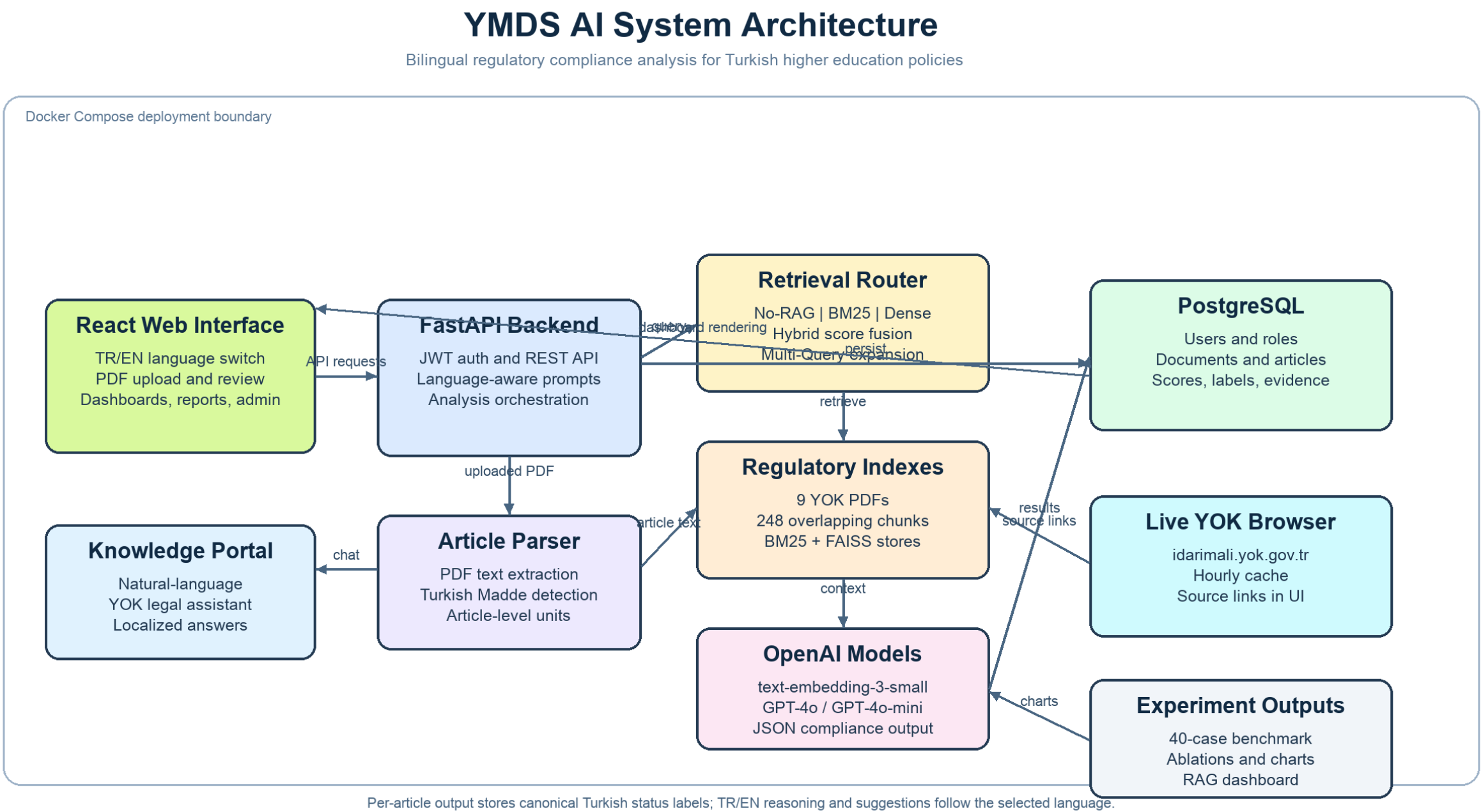
Limitation

Non-compliant recall remains challenging; human review is required.

Future Work

Automatic re-indexing, larger expert benchmark, better validation calibration.

Architecture



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